



OSTEOPOROSIS DETECTION

Using Deep Learning Techniques

SPECIFICATION OF REFERENCES

- This project is supported by several research studies related to deep learning, medical image analysis, and osteoporosis diagnosis.

This project is based on several research works related to osteoporosis detection using deep learning and medical image analysis.

Important research contributions related to this work include:

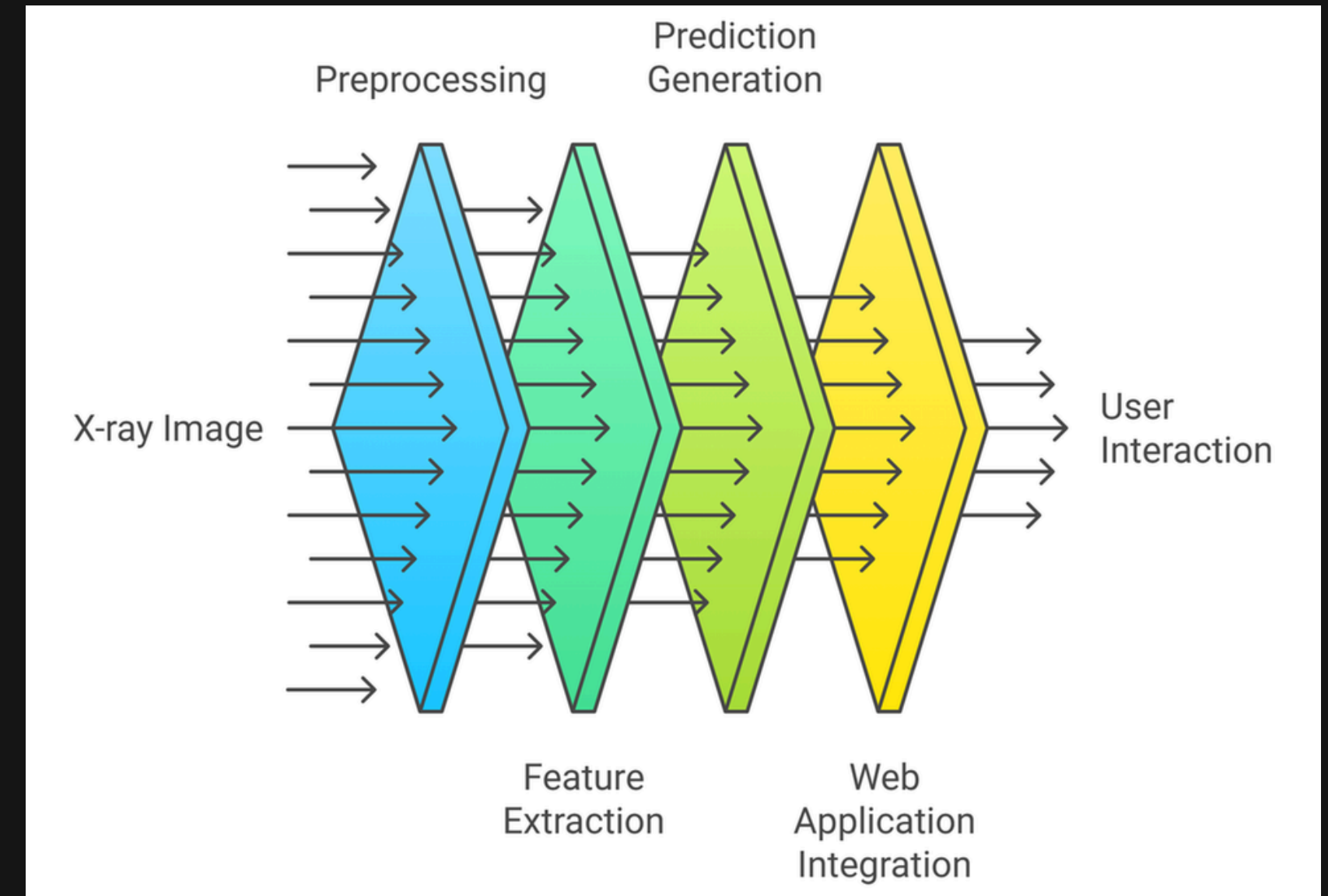
- Deep learning approaches for osteoporosis detection using medical images
- CNN-based classification of knee X-ray images for bone disease detection
- Transfer learning models such as VGG16, VGG19, ResNet50, InceptionV3, and Xception
- Explainable AI techniques like Grad-CAM for visualizing important regions in medical images
- AI-based automated systems for assisting early diagnosis of bone diseases

These studies motivated the development of our deep learning-based osteoporosis detection system.

These research works demonstrate the effectiveness of AI-driven automated screening systems and motivated the development of our deep learning framework for multiclass osteoporosis classification using knee X-ray images.

DESCRIPTION OF CONCEPTS AND TECHNICAL DETAILS

- The proposed system uses Deep Learning and Transfer Learning techniques to automatically classify knee X-ray images into three categories:
 - Normal
 - Osteopenia
 - Osteoporosis
- The system analyzes bone patterns and density changes present in knee radiographs to identify osteoporosis conditions.
- The framework helps in early detection and decision support for clinicians.



DEEP LEARNING MODELS USED

- Transfer learning allows models pre-trained on large datasets to be adapted for medical imaging tasks with limited data.
- The Custom CNN architecture was specifically designed for knee X-ray analysis.

Model	Architecture Type	Key Strength	Performance Trend
VGG16	Deep CNN (16 layers)	Simple & stable architecture	Good baseline accuracy
ResNet50	Residual Network	Handles deep networks efficiently using skip connections	Better than VGG
InceptionV3	Multi-scale CNN	Captures fine texture and spatial details	High accuracy
VGG19	Deep CNN (19 layers)	Rich feature extraction	Comparable / slightly better than VGG16
DenseNet121	Dense Connected Network	Reuses features from previous layers, improves gradient flow	Strong performance in medical imaging
MobileNetV2	Lightweight CNN	Designed for mobile & real-time applications	Slightly lower but efficient accuracy
Xception	Depthwise Separable CNN	Efficient spatial feature extraction with fewer parameters	Near Inception-level accuracy
Custom CNN	Domain-specific CNN	Tailored for knee X-ray patterns	Highest performance on our dataset

Table 1: Comparison of Models

DATA PREPROCESSING

- To improve model performance, several preprocessing techniques were applied:
 - Image resizing to 224×224 resolution
 - Pixel value normalization
- Data augmentation techniques:
 - Rotation
 - Horizontal flipping
 - Zooming
- These steps help:
 - Reduce overfitting
 - Improve model generalisation.
 - Handle dataset imbalance.
- These preprocessing steps are essential for training reliable deep learning models on medical images.

UI APPLICATION

Model Configuration

Choose Model Architecture ⓘ

InceptionV3

Model Information

Settings

Show Image Preprocessing

Show Detailed Confidence

Show Prediction History

Clear History

Osteoporosis Classification AI

Advanced Deep Learning System for Bone Health Analysis

Upload X-Ray Image

Choose an X-ray image...

Drag and drop file here

Limit 200MB per file • PNG, JPG, JPEG

Browse files

Classification Results

Please upload an X-ray image to start classification

Normal Bone

Healthy bone density with no signs of deterioration. Regular bone density is maintained.

Osteopenia

Lower than normal bone density but not severe enough to be classified as osteoporosis.

Osteoporosis

Significant bone density loss, making bones fragile and more likely to fracture.

Osteoporosis Classification AI | Powered by Deep Learning

⚠️ This tool is for research purposes only. Please consult healthcare professionals for medical diagnosis.

CONCLUSION & DISCUSSION

- The project successfully developed a deep learning-based system for multiclass osteoporosis detection using knee X-ray images.
- Multiple CNN architectures were trained and evaluated.
- Among all models:
 - Custom CNN achieved the highest accuracy of 89%.
 - InceptionV3 achieved 88% accuracy.
 - Xception achieved 87% accuracy.

Grad-CAM explainability techniques were used to visualise the regions influencing model predictions.

The results demonstrate that deep learning can effectively support automated osteoporosis screening and clinical decision support.

FUTURE SCOPE

- Develop a more interactive web or mobile application to allow doctors and healthcare workers to easily upload X-ray images and obtain predictions.

1. User-Friendly Interface

- The current work can be extended into a full research publication to contribute to the field of AI-assisted medical diagnosis.

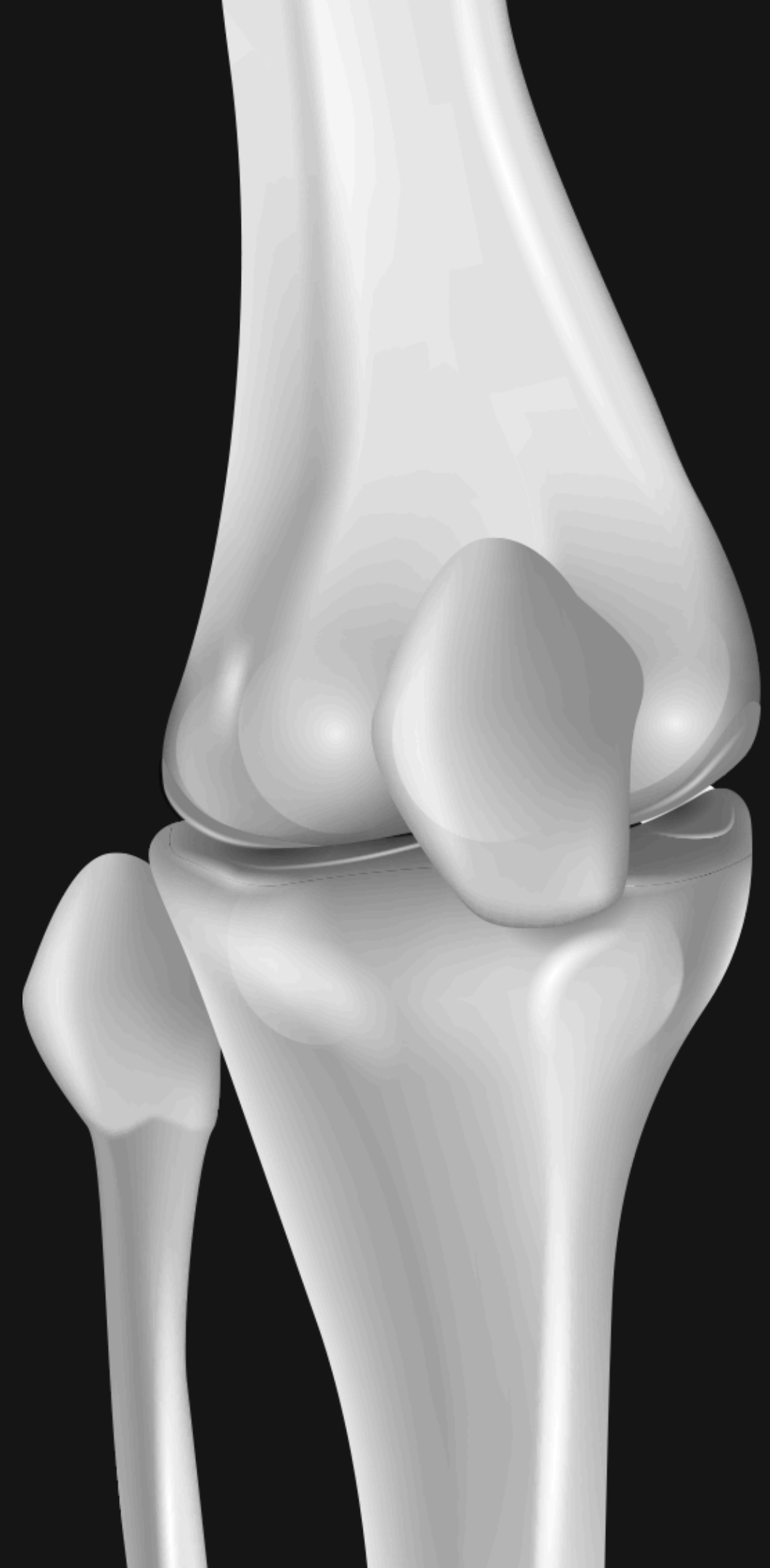
2. Research Paper Publication

- Collect larger knee X-ray datasets from Indian hospitals to improve model accuracy and ensure better generalization across different populations.

3. Dataset Collection from India

- Combine predictions from multiple deep learning models using ensemble techniques to further improve classification accuracy and robustness.

4. Integration of Ensemble Models



THANK YOU